LINUX PROGRAMMING : ASSIGNMENT-3

1. Distinguish between man and whatis commands? Justify with proper example.

ANS man Command (Manual)
* Purpose: Provides the full, detailed manual page for a specific command or topic.
* Output: Displays comprehensive documentation, including the command's name, synopsis, detailed description, options, return values, and related files.
* Functionality: Used when the user knows the command name and needs in-depth technical usage information.
* Equivalent Flag: Can be executed using the man command itself (e.g., man ls).

whatis Command
* Purpose: Displays a concise, one-line description of a specific command.
* Output: Shows only the command name and a single-sentence summary of its function (e.g., ls (1) - list directory contents).
* Functionality: Used for a quick reference to remind the user what a known command does.
* Equivalent Flag: Is equivalent to running the manual command with the -f flag (i.e., man -f).

2. Use the tee command to save the output of ls -l into a file while also displaying it.

ANS Pavan: cite_start]The tee command is necessary to divert a data stream to both a file and the standard output simultaneously.
Command:
Pavan: ls -l | tee file_contents_list.txt
Pavan: [cite_start]The detailed directory listing generated by ls -l is piped (|) to tee[cite_start], which duplicates the output stream. [cite_start]One copy goes to the terminal screen, and the other is saved to file_contents_list.txt.

3. Explain with an example how the tee command can be used in logging.

ANS [cite_start]tee is crucial in logging because it enables real-time monitoring of a process while ensuring every step and message is securely recorded. This prevents the loss of crucial output data if the system were to crash during execution.
Logging Example:
To document the process of installing a new package, where output needs to be reviewed immediately and kept for historical record:

sudo apt install my_app | tee -a installation_audit.log

[cite_start]The output of the installation is printed to the terminal, allowing the administrator to watch for errors. [cite_start]Simultaneously, the complete, timestamped log is appended (-a) to installation_audit.log.

4. List the steps involved in installing Ubuntu 25.04 LTS on Oracle VirtualBox.

ANS 1. [cite_start]Preparation: Download and install Oracle VirtualBox on the host machine. [cite_start]Obtain the Ubuntu 25.04 LTS installation media (ISO file).

2. [cite_start]VM Creation: In VirtualBox, use the "New" wizard to create a virtual machine (VM), selecting Linux and the appropriate Ubuntu version
.
3. [cite_start]Resource Allocation: Assign necessary resources (e.g., RAM and CPU cores) and create a virtual hard disk (VDI recommended) of sufficient size.

4. [cite_start]Media Mounting: In the VM's settings, locate the storage controller and "mount" the downloaded Ubuntu ISO file as the virtual CD/DVD drive.

5. [cite_start]Execution & Installation: Start the VM, which boots from the ISO. [cite_start]Choose the "Install Ubuntu" option and follow the graphical setup wizard, configuring language, time zone, disk partitioning (usually "Erase disk and install" for a new VM), and creating the primary user account.

6. [cite_start]Completion: Wait for the installation to finish and reboot the VM to use the newly installed operating system.

5.During Ubuntu OS installation, you face a Kernel Panic Error. How would you troubleshoot it?

ANS [cite_start]A Kernel Panic signifies a deep system error where the core operating system cannot recover. Troubleshooting focuses on isolating hardware or media issues in the virtual environment.
1. [cite_start]Verify ISO Integrity: The installation source might be corrupted. Re-download the Ubuntu ISO and confirm its integrity using a checksum utility.
2. [cite_start]Adjust Virtualization Settings: Ensure that hardware virtualization (VT-x/AMD-V) is correctly enabled in the host machine's BIOS/UEFI and that the VM's settings (under System \rightarrow Acceleration) reflect this.
3. [cite_start]Reduce VM Resources: Temporarily reduce the allocated RAM or CPU cores to rule out conflicts caused by resource over-allocation or issues related to a specific hardware configuration.
4. Use Safe Boot Options: Restart the installation and, at the GRUB menu, try selecting the "Safe Graphics" or a similar recovery mode option. [cite_start]This may bypass display driver issues that can trigger a panic.

6.Write the command to display the system’s hostname? How to change hostname using sysctl command?

ANS Pavan: cite_start]Display Hostname: The current name of the machine on the network is shown by the hostname command.
Pavan: hostname
] Pavan: cite_start]Changing Hostname using sysctl: The sysctl command manages kernel parameters at runtime, including the kernel.hostname variable. While it can temporarily change the hostname, this change is lost upon reboot.

Pavan: sudo sysctl kernel.hostname="TempHostName

7.Which command is used to show the calendar of the year 1984 with August month?

ANS Pavan: cal 8 1984
Pavan: [cite_start]This specifies month 8 (August) and the year 1984.

8. Write a command to display system uptime and logged-in users together.

ANS Pavan: [cite_start]The w command provides a consolidated view that includes the system's uptime, load average, and a list of users currently logged in with their corresponding activities.
Pavan: w
Pavan: * [cite_start]Alternative: You can manually combine the output of two specific commands using a pipeline: uptime && who.

9. Use the find command to list all ".c" files in /home/user.

ANS   Pavan: [cite_start]The find command is used to traverse a file hierarchy and locate files based on specific criteria.
Pavan: find /home/user -name "*.c"
Pavan: * [cite_start]/home/user is the starting directory.
* [cite_start]-name "*.c" restricts the search to files whose names end in the .c extension.

10. How do you change file permissions to allow only the owner to read and write?

ANS   Pavan: To enforce that only the file owner can read and modify a file, all other access permissions (for group and others) must be removed. This corresponds to the numeric code 600 (Read=4, Write=2, Execute=0).
] Pavan: chmod 600 specific_file
Pavan: chmod u=rw,go= specific_file